

Vector Databases and High-Dimensional Indexing

Purpose of Vector Databases Vector databases store, index, and query high-dimensional embeddings generated by machine learning models. They enable rapid nearest-neighbor search for semantic comparison.

Similarity Metrics Common distance metrics to measure semantic closeness include:

- **Cosine Similarity:** Measures the angle between vectors, ignoring magnitude.
- **Euclidean Distance (L2):** Measures the straight-line distance between points in space.
- **Dot Product:** Measures magnitude and direction alignment.

Indexing Algorithms To query millions of vectors in milliseconds, databases use Approximate Nearest Neighbor (ANN) index structures:

- **FAISS:** Facebook AI Similarity Search library, optimized for local dense vector search in RAM.
- **HNSW:** Hierarchical Navigable Small World, creating multi-layer graphs for rapid traversal.
- **IVF:** Inverted File Index, clustering vector spaces to restrict queries to subset partitions.